
VEHICLE DETECTION IN REAL TIME ENVIRONMENT DURING FOG/HAZY DAY USING MATLAB

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Submitted in the partial Fulfilment of the requirements for the award the

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**BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

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CERTIFICATE

This is to certify that the project entitled “ **VEHICLE DETECTION IN REAL TIME ENVIRONMENT DURING FOG/HAZY DAY USING MATLAB**” is the bonafied work done by BANDARI HARI VISHNU (23R21A0408), BOGGULA NANDINI (23R21A0414), GUNTURI PRIYA DARSHINI (23R21A0426), NAYAKAVADI SHRUTHI (23R21A0444) in partial fulfilment of the requirement for the award of the degree of B.Tech in Electronics and Communication Engineering, during the academic year 2025-26.

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ABSTRACT

Fog and hazy weather conditions significantly reduce road visibility and create major challenges for drivers and traffic monitoring systems. Due to the presence of fog, images captured by cameras suffer from low contrast, poor brightness, and loss of important visual details, which makes vehicle detection difficult using traditional techniques. Therefore, developing an effective system for detecting vehicles in foggy environments is important for improving road safety and traffic management. This project focuses on vehicle detection in a real-time foggy or hazy environment using MATLAB by applying image processing techniques to enhance degraded images.

The proposed system first improves the quality of foggy images using enhancement methods such as Contrast Limited Adaptive Histogram Equalization (CLAHE), histogram equalization, adaptive gamma correction, and image sharpening to increase visibility and restore important image details. After the enhancement process, vehicle detection techniques are applied to identify and highlight vehicles present in the image or video frames. The system is implemented using MATLAB with the Image Processing Toolbox, which allows efficient processing and analysis of visual data.

Overall, this project aims to improve vehicle detection accuracy under low-visibility conditions and contribute to applications such as traffic monitoring, intelligent transportation systems, and driver assistance systems, thereby enhancing road safety during foggy weather conditions.

CHAPTER 1

INTRODUCTION

In modern transportation systems, safety and efficiency are highly dependent on the ability to detect vehicles and obstacles accurately in real time. Vehicle detection plays a crucial role in applications such as intelligent transportation systems, driver assistance systems, autonomous vehicles, traffic monitoring, and surveillance systems. However, environmental factors such as fog, haze, dust, and smoke significantly degrade the quality of captured images or videos, making object detection difficult.



Fig. 1.1 Images (a) and (b) are captured in a foggy weather

Foggy or hazy weather conditions reduce visibility due to the scattering and absorption of light by tiny particles present in the atmosphere. As a result, images captured under these conditions suffer from poor contrast, reduced color fidelity, blurred details, and low brightness. These degradations make it difficult for computer vision systems to accurately detect vehicles in real time.

Vehicle detection in foggy conditions is particularly important for road safety. Many road accidents occur due to reduced visibility during foggy weather, especially on highways and rural roads. Drivers often fail to notice nearby vehicles, pedestrians, or obstacles until it is too late. Therefore, developing intelligent systems capable of detecting vehicles in such environments is essential for improving road safety and traffic management.

To address this issue, image enhancement and image processing techniques can be applied before performing object detection. These techniques improve the visibility of captured images by removing noise, enhancing contrast, and restoring important details. MATLAB provides powerful tools and libraries for implementing digital image processing algorithms that can enhance foggy images and detect vehicles effectively.

The main objective of this project is to develop a system that can detect vehicles in real time under foggy or hazy environmental conditions using MATLAB. The system first enhances the visibility of the captured image using image processing techniques and then identifies vehicles using detection algorithms. By improving image clarity and contrast, the proposed system ensures reliable vehicle detection even in low visibility environments.

1.1 DIGITAL IMAGE PROCESSING

Digital Image Processing (DIP) is the process of manipulating digital images using computer algorithms to improve image quality or extract useful information. A digital image is a representation of a two-dimensional image as a set of digital values called pixels.

In simple terms, digital image processing involves performing operations on images such as filtering, enhancement, segmentation, and feature extraction. These operations allow computers to analyze and interpret visual information.

1.1.2 Digital Image Acquisition

Digital image acquisition is the first step in any computer vision system. It involves capturing an image of a scene using a camera and converting it into a digital format that can be processed by a computer.

A digital camera consists of the following components:

1. Optical lens
2. Image sensor (CCD or CMOS)
3. Analog-to-Digital Converter (ADC)
4. Image processing unit

The process of converting a real-world scene into a digital image involves two main steps:

1. Sampling

Sampling converts a continuous image into discrete spatial coordinates.

2. Quantization

Quantization converts continuous intensity values into discrete digital values.

After these two steps, the image becomes a digital image composed of pixels.

1.1.3 Mathematical Representation of Digital Images

A digital image can be represented as a two-dimensional function:

$$f(x, y)$$

Where:

- x = horizontal spatial coordinate
- y = vertical spatial coordinate
- $f(x, y)$ = intensity value at that point

If the values of x , y , and $f(x, y)$ are discrete, the image is considered a digital image.

1.1.4 Image as Matrix

In MATLAB, images are represented as matrices.

For example, a grayscale image is represented as:

$$f = \begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \dots & \dots & \dots & \dots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$

Where:

- M represents rows
- N represents columns

Each element of the matrix corresponds to a pixel value.

1.1.5 Grayscale Image

A grayscale image contains only intensity information and does not include color. Each pixel value represents brightness and usually ranges from:

0 → Black

255 → White

Grayscale images are commonly used in image processing because they require less computational complexity compared to color images.

1.1.6 Color Image

A color image consists of three main channels:

- Red (R)
- Green (G)
- Blue (B)

Each pixel contains three values representing the intensity of these colors. These channels combine to produce a full color image.

Color images provide more visual information but require more processing power.

1.1.7 Image Formation Model in Foggy Environment

When images are captured in foggy environments, the intensity received by the camera is affected by atmospheric scattering.

The fog imaging model is expressed as:

$$I(x) = J(x)t(x) + A(1 - t(x))$$

Where:

- $I(x)$ = observed foggy image
- $J(x)$ = actual scene radiance
- A = atmospheric light
- $t(x)$ = transmission map

The transmission map represents the amount of light that reaches the camera.

It is defined as:

$$t(x) = e^{-\beta d(x)}$$

Where:

- β = atmospheric scattering coefficient
- $d(x)$ = distance between object and camera

Interpretation

When distance increases:

$$d(x) \uparrow \Rightarrow t(x) \downarrow$$

Thus the captured image becomes more foggy and low contrast.

1.1.8 Reading Images:

Images are read into the MATLAB environment using function `imread()` whose syntax is `imread('filename')`

Fig.1 Reading images

Format name	Description	recognized extension
TIFF	Tagged Image File Format	.tif, .tiff
JPEG	Joint Photograph Experts Group	.jpg, .jpeg
GIF	Graphics Interchange Format	.gif
BMP	Windows Bitmap	.bmp
PNG	Portable Network Graphics	.png

In MATLAB, images are read using the built-in function `imread()`, which loads an image file from a specified location and stores it in a matrix variable for further processing. The filename is given as a string that contains the complete name of the image file along with its extension. For example, the command `f = imread('image.jpg');` reads a JPEG image file named *image.jpg* and stores it in the image array `f`.

The `size(f)` function returns the number of rows and columns of the image, while `[M,N] = size(f)` assigns these values to variables representing the image dimensions. For instance, an image may be stored as a `uint8` array, meaning that each pixel value occupies one byte of memory and ranges from 0 to 255. These functions are useful when analyzing and processing images in MATLAB, especially in applications such as vehicle detection in foggy or hazy environments, where understanding the image size and structure is essential for further processing.

1.1.9 Displaying Images:

Pictures are shown on the MATLAB work area utilizing capacity `imshow`, which has the fundamental language structure:

```
imshow(f,g)
```

Where `f` is a picture exhibit, and `g` is the quantity of force levels used to show it. In the event that `g` is excluded, it defaults to 256 levels. Utilizing the language structure

```
imshow(f, {low high})
```

Shows as dark all qualities not exactly or equivalent to low and as white all qualities more prominent than or equivalent to high. The qualities in the middle of are shown as halfway force esteems utilizing the default number of levels. Finally the language structure

```
imshow(f,[ ]) 
```

Sets variable low to the base estimation of exhibit `f` and high to its most extreme esteem. This type of `imshow` is helpful for showing pictures that have a low powerful range or that have positive and negative qualities.

Images are displayed using the `imshow()` function, which is commonly used in image processing to visualize digital images stored in matrix form. After reading an image using the `imread()` function, it can be displayed on the screen using the command `imshow(f)`, where `f` represents the image matrix.

For example, the command `figure, imshow(g)` opens a new figure window and displays the image `g` without replacing the previously displayed image. Multiple commands can also be written in a single line separated by commas or semicolons. If an image appears too dark or has a low intensity range, MATLAB allows adjusting the display scaling

using `imshow(h, [])`. This command automatically stretches the intensity values of the image for better visualization. These display functions are important in image processing applications such as vehicle detection in foggy or hazy environments, where enhanced images and detection results must be visualized clearly for analysis and verification.

1.1.10 WRITING IMAGES

Pictures are composed to circle utilizing capacity `imwrite`, which has the accompanying fundamental language structure:

`Imwrite (f,'filename')`

With this sentence structure, the string contained in `filename` must incorporate a perceived record design expansion .Alternatively, the coverted configuration can be indicated expressly with a third info contention. `>>imwrite(f,'image','tif')`

For instance, the accompanying direction composes `f` to a TIFF record named `patient10_run1`:

`>>imwrite(f,'image.tif')`

On the off chance that `filename` contains no way data, at that point `imwrite` spares the record in the present working index.

Images are written to storage using the `imwrite()` function. The basic syntax of this function is `imwrite(f,'filename')`, where `f` represents the image matrix and *filename* specifies the name of the file along with a valid image format extension. For example, the command `imwrite(f,'image.tif')` saves the processed image `f` as a TIFF file named *image.tif*. The `imwrite()` function supports several image formats such as JPEG, PNG, BMP, and TIFF, and it allows additional parameters depending on the selected format. For JPEG images, a quality parameter can be specified to control the level of compression. The syntax for this is `imwrite(f,'filename.jpg','quality',q)`, where `q` is a value between 0 and 100. Lower values produce higher compression but reduce image quality, while higher values maintain better image clarity.

After enhancing foggy images or detecting vehicles using MATLAB algorithms, the results can be stored using `imwrite()` for further evaluation or documentation. Proper selection of image format, compression level, and resolution ensures that the stored

images maintain sufficient quality while optimizing storage space. This capability is particularly useful when handling large numbers of road images captured in real-time traffic monitoring systems or when generating outputs for research reports and experimental analysis.

1.2 IMAGE ENHANCEMENT

Image enhancement is basically improving the Interpretability or perception of information in images for human viewers and providing better input for other automated image processing techniques. There exist many techniques that can enhance a digital image without spoiling it. The enhancement methods can broadly be divided in to the following two categories:

1. Spatial Domain Methods.
2. Frequency Domain Methods.

In spatial domain techniques, we directly deal with the image pixels. The pixel values are manipulated to achieve desired enhancement. Image enhancement is applied in every field where images are ought to be understood and analysed. For example, medical image analysis, analysis of images from satellites etc. In this section we briefly describe the various image enhancement techniques.

1.2.1 Spatial domain methods

The value of a pixel with coordinates (x, y) in the enhanced image $\hat{F}$ is the result of performing some operation on the pixels in the neighbourhood of (x, y) in the input image, F . Neighbourhoods can be any shape, but usually they are rectangular.

1.2.2. Grey scale manipulation

The simplest form of operation is when the operator T only acts on a 1×1 pixel neighbourhood in the input image, that is $\hat{F}(x, y)$ only depends on the value of F at (x, y) . This is a grey scale transformation or mapping. The simplest case is thresholding where the intensity profile is replaced by a step function, active at a chosen threshold value. In this case any pixel with a grey level below the threshold in the input image gets mapped to 0 in the output image. Other pixels are mapped to 255. Other grey scale transformations are outlined in figure 3.1 below.

1.2.3. Histogram Equalization

Histogram equalization is a common technique for enhancing the appearance of images. Suppose we have an image which is predominantly dark. Then its histogram would be skewed towards the lower end of the grey scale and all the image detail is compressed into the dark end of the histogram. If we could 'stretch out' the grey levels at the dark end to produce a more uniformly distributed histogram then the image would become much clearer.

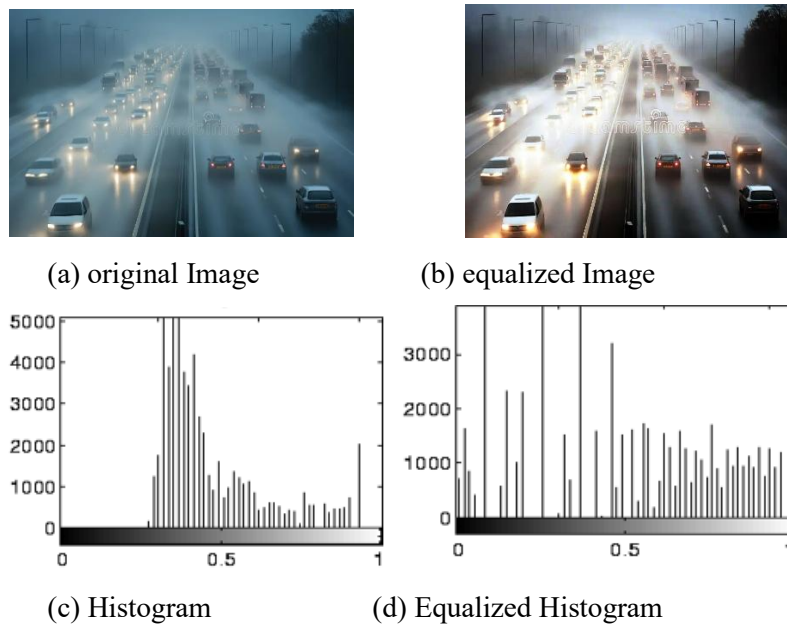


Fig.1.2. The original image and its histogram, and the equalized versions. Both images are quantized to 64 grey levels.

Histogram equalization involves finding a grey scale transformation function that creates an output image with a uniform histogram (or nearly so).

How do we determine this grey scale transformation function? Assume our grey levels are continuous and have been normalized to lie between 0 and 1. We must find a transformation T that maps grey values r in the input image F to grey values $s = T(r)$ in the transformed image $\hat{F}$.

It is assumed that

- T is single valued and monotonically increasing, and $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$.
- The inverse transformation from s to r is given by $r = T^{-1}(s)$.

If one takes the histogram for the input image and normalizes it so that the area under the histogram is 1, we have a probability distribution for grey levels in the input image $P_r(r)$. If we transform the input image to get $s = T(r)$ what is the probability distribution $P_s(s)$?

From probability theory it turns out that

$$P_s(s) = P_r(r) \frac{dr}{ds'} \quad (1.1)$$

where $r = T^{-1}(s)$.

Consider the transformation $s = T(r) = \int_0^r P_r(w)dw$ (1.2)

This is the cumulative distribution function of r . Using this definition of T we see that the derivative of s with respect to r is

$$\frac{ds}{dr} = P_r(r) \quad (1.3)$$

Substituting this back into the expression for P_s , we get

$$P_s(s) = P_r(r) \frac{1}{P_r(r)} = 1 \quad (1.4)$$

for all s , where $0 \leq s \leq 1$. Thus, $P_s(s)$ is now a uniform distribution function, which is what we want.

1.2.5. Image Smoothing

The aim of image smoothing is to diminish the effects of camera noise, spurious pixel values, missing pixel values etc. There are many different techniques for image smoothing; we will consider neighbourhood averaging and edge-preserving smoothing.

1.2.6. Edge preserving smoothing

Neighbourhood averaging or Gaussian smoothing will tend to blur edges because the high frequencies in the image are attenuated. An alternative approach is to use median filtering. Here we set the grey level to be the median of the pixel values in the neighbourhood of that pixel. The median m of a set of values is such that half the values in the set are less than m and half are greater. For example, suppose the pixel values in a 3×3 neighbourhood are (10, 20, 20, 15, 20, 20, 20, 25, and 100). If we sort the values we get (10, 15, 20, 20, |20|, 20, 20, 25, 100) and the median here is 20. The outcome of

median filtering is that pixels with outlying values are forced to become more like their neighbours, but at the same time edges are preserved. Of course, median filters are non-linear.

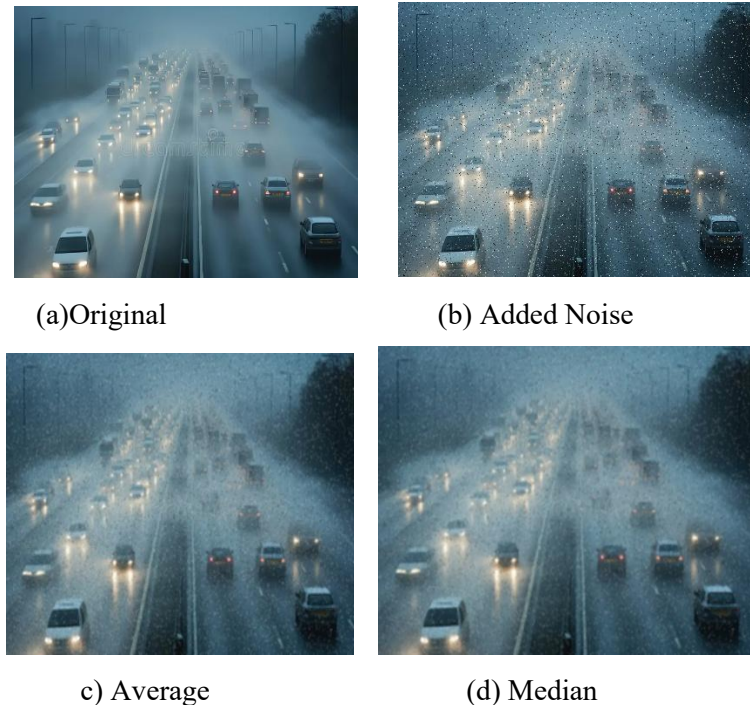


Fig. 1.3. Image of traffic roads; with salt and pepper noise; the result of averaging; and the result of median filtering.

Median filtering is in fact a morphological operation. When we erode an image, pixel values are replaced with the smallest value in the neighbourhood. Dilating an image corresponds to replacing pixel values with the largest value in the neighbourhood. Median filtering replaces pixels with the median value in the neighbourhood. It is the rank of the value of the pixel used in the neighbourhood that determines the type of morphological operation.

1.2.7. Image sharpening

The main aim in image sharpening is to highlight fine detail in the image, or to enhance detail that has been blurred (perhaps due to noise or other effects, such as motion). With image sharpening, we want to enhance the high-frequency components; this implies a spatial filter shape that has a high positive component at the centre (see figure 3.4 below).

A simple spatial filter that achieves image sharpening is given by

$$\begin{matrix} -1/9 & -1/9 & -1/9 \\ & & \end{matrix}$$

$$\begin{matrix} -1/9 & 8/9 & -1/9 \\ & & \end{matrix}$$

$$\begin{matrix} -1/9 & -1/9 & -1/9 \\ & & \end{matrix}$$

Since the sum of all the weights is zero, the resulting signal will have a zero DC value (that is, the average signal value, or the coefficient of the zero-frequency term in the Fourier expansion). For display purposes, we might want to add an offset to keep the result in the $[0, 255]$ range.

1.2.8. Frequency domain methods

In frequency domain methods, the image is first transferred into frequency domain. It means that, the Fourier Transform of the image is computed first. All the enhancement operations are performed on the Fourier transform of the image and then the Inverse Fourier transform is performed to get the resultant image.

The idea of blurring an image by reducing its high frequency components or sharpening an image by increasing the magnitude of its high frequency components is intuitively easy to understand. However, computationally, it is often more efficient to implement these operations as convolutions by small spatial filters in the spatial domain. Understanding frequency domain concepts is important and leads to enhancement techniques that might not have been thought of by restricting attention to the spatial domain.

1.2.9. Filtering

Low pass filtering involves the elimination of the high frequency components in the image. It results in blurring of the image (and thus a reduction in sharp transitions associated with noise). An ideal low pass filter would retain all the low frequency components and eliminate all the high frequency components. However, ideal filters suffer from two problems: blurring and ringing. These problems are caused by the shape of the associated spatial domain filter, which has many undulations. Smoother transitions in the frequency domain filter, such as the Butterworth filter, achieve much better results.

1.2.10 Homomorphism filtering

In foggy or hazy environments, captured road images often suffer from uneven illumination, low contrast, and multiplicative noise caused by fog particles scattering light. To improve the visibility of objects such as vehicles, homomorphic filtering is used as an image enhancement technique. An image $F(x, y)$ can be modeled as the product of two components: the illumination component $i(x, y)$, which represents the amount of light falling on the scene, and the reflectance component $r(x, y)$, which represents the light reflected from objects. This relationship is expressed as

$$F(x, y) = i(x, y) r(x, y)$$

where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$. In foggy road images, illumination varies slowly across the image and appears as low-frequency components, while reflectance corresponds to object details such as vehicle edges and textures, which appear as high-frequency components. Since these components are multiplied together, they cannot be directly separated using the Fourier transform.

To make the components separable, the logarithm of the image intensity is taken. Let

$$z(x, y) = \ln F(x, y) = \ln i(x, y) + \ln r(x, y)$$

Taking the Fourier transform of this equation converts the multiplicative relationship into an additive one:

$$Z(\omega, v) = I(\omega, v) + R(\omega, v)$$

where $Z(\omega, v)$, $I(\omega, v)$, and $R(\omega, v)$ are the Fourier transforms of $z(x, y)$, $\ln i(x, y)$, and $\ln r(x, y)$ respectively. At this stage, a filter with transfer function $H(\omega, v)$ is applied to suppress the low-frequency illumination component and enhance the high-frequency reflectance component:

$$S(\omega, v) = H(\omega, v) Z(\omega, v)$$

The enhanced image in the spatial domain is then obtained by applying the inverse Fourier transform:

$$s(x, y) = F^{-1}[S(\omega, v)]$$

Finally, since the logarithm was applied earlier, the exponential operation is used to obtain the enhanced image:

$$\hat{F}(x, y) = \exp [s(x, y)]$$

As a result, the processed image provides clearer edges and structures, which helps subsequent steps such as edge detection, segmentation, and vehicle identification perform more accurately in low-visibility conditions.

Thus, the process of homomorphism filtering can be summarized by the following diagram:

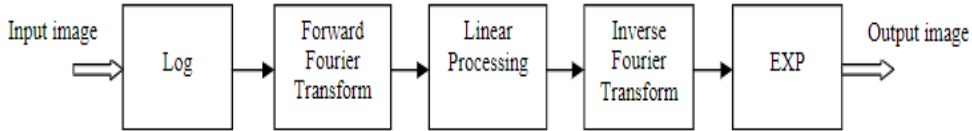


Fig 1 block diagram of homomorphism filtering

1.3 EDGE DETECTION

Edge detection is an important step in the vehicle detection system because it helps identify the boundaries of objects such as vehicles, road markings, and obstacles in the enhanced foggy image. Edges represent points where there is a significant change in pixel intensity, which usually corresponds to object borders. In foggy conditions, vehicle boundaries become less visible due to reduced contrast, so detecting edges after image enhancement helps in identifying the structure and shape of vehicles. Edge detection methods generally rely on calculating the image gradient, which measures the rate of change of intensity in horizontal and vertical directions.

1.3.1 Gradient Computation

The first step in edge detection is computing the gradient of the image intensity function. The gradient indicates the direction and magnitude of the greatest change in intensity. If the image is represented as $f(x, y)$, the gradient vector is given by

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Where:

$\frac{\partial f}{\partial x}$ represents the change of intensity in the horizontal direction, and

$\frac{\partial f}{\partial y}$ represents the change in the vertical direction.

The magnitude of the gradient is calculated as $|\nabla f| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$

A large gradient magnitude indicates the presence of an edge in the image. In the foggy vehicle detection project, this gradient computation helps highlight vehicle contours and structural boundaries.

1.3.2. Edge Detection Operators

To compute the gradient in discrete images, edge detection operators are used. These operators apply convolution masks to estimate intensity changes. Common operator used is Canny filters.

1.3.3 Canny Edge Detection

The Canny edge detector is widely used because it provides accurate edge detection while suppressing noise. It works through several stages:

Gaussian smoothing to reduce noise ,Gradient computation ,Non-maximum suppression to thin edges ,Double thresholding and edge tracking.

Mathematically, the gradient magnitude used in Canny detection is

$$G = \sqrt{G_x^2 + G_y^2}$$

and the edge direction is $\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$

1.3.4 Edge Map Generation

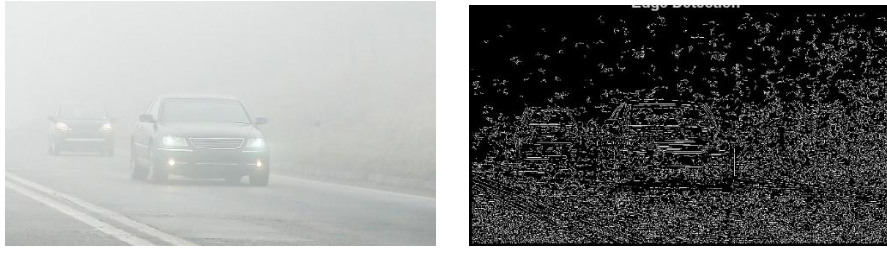
The final result of the edge detection process is an edge map, which is a binary image that highlights the locations of edges in the original image. In this map, pixels corresponding to edges are assigned a value of 1, while all other pixels are assigned 0.

$$E(x, y) = \begin{cases} 1, & \text{if edge is detected} \\ 0, & \text{otherwise} \end{cases}$$

The edge map reveals the shape and structure of vehicles, making it easier to perform segmentation and object detection.

In the foggy vehicle detection project, this edge map plays a critical role in identifying the outlines of vehicles and separating them from the background. Once the edges are

detected, further steps such as segmentation, morphological processing, and connected component analysis can be applied to accurately detect and mark vehicles in the image.



(a) Original Image

(b) Edge Detection

Fig. 1.4. Original image and Edge Detection

1.4 IMAGE SEGMENTATION

Image segmentation is the process of dividing an image into meaningful regions so that important objects can be separated from the background. In the foggy vehicle detection project, segmentation helps isolate vehicle regions from the road and surrounding environment after the edge detection stage. Since fog reduces contrast and visibility, segmentation plays a crucial role in identifying potential vehicle areas by grouping pixels with similar intensity values. One of the most common techniques used for segmentation is thresholding, where pixels are classified into object or background categories based on their intensity.

1.4.1 Thresholding

Thresholding is a simple yet effective segmentation technique used to convert a grayscale image into a binary image. The method compares each pixel intensity with a predefined threshold value T . If the pixel value is greater than the threshold, it is considered part of the object; otherwise, it is treated as background.

Mathematically, thresholding can be expressed as:

$$B(x, y) = \begin{cases} 1, & \text{if } f(x, y) > T \\ 0, & \text{if } f(x, y) \leq T \end{cases}$$

where

$f(x, y)$ = intensity value of the input image, T = threshold value, $B(x, y)$ = output binary image

In the vehicle detection system, pixels belonging to vehicles often have different intensity values compared to the road background. Thresholding helps highlight these regions and separate vehicles from the

1.4.2 Otsu's Method

To overcome the limitations of manual threshold selection, Otsu's method is commonly used. This technique automatically calculates an optimal threshold value by analyzing the histogram of the image and minimizing the intra-class variance between foreground and background pixels.

Let the image contain two classes:

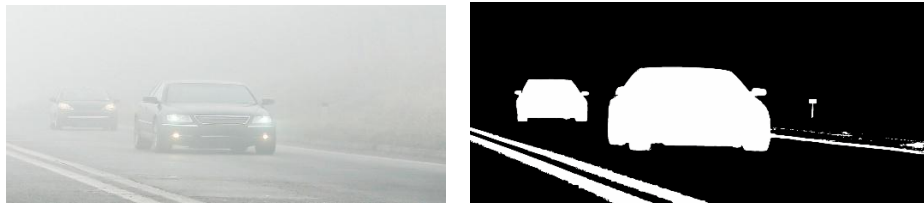
- Background pixels , Object pixels

Otsu's method determines the threshold T that minimizes the weighted variance:

$$\sigma_w^2(T) = w_0(T)\sigma_0^2(T) + w_1(T)\sigma_1^2(T)$$

where

w_0, w_1 = probabilities of the two classes , σ_1^2 = variances of the classes.



(a) Original Image

(b) Image Segmentation

Fig. 1.5. Original image and Image Segmentation

1.5 MORPHOLOGICAL PROCESSING

Morphological processing is applied to the segmented binary image to refine the detected vehicle regions and remove unwanted noise. After image segmentation, the binary image may contain small isolated pixels, gaps, and irregular boundaries due to fog, illumination variations, and sensor noise. Morphological operations analyze the shape and structure of objects using a structuring element and modify the image accordingly. In this project, morphological processing helps improve the shape of detected vehicle regions, making them more suitable for the final vehicle detection stage.

1.5.1 Erosion

Erosion is used to remove small noise pixels and shrink object boundaries. This operation eliminates isolated pixels that do not belong to the main vehicle regions.

Mathematically, erosion is defined as

$$A \ominus B = \{z \mid B_z \subseteq A\}$$

where

A = input binary image , B = structuring element

1.5.2. Dilation

Dilation is used to expand object boundaries and restore parts of objects that may have been removed during erosion.

Mathematically,

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\}$$

In dilation, a pixel becomes **1** if at least one pixel under the structuring element is **1**.

1.5.3. Opening

Opening is the combination of erosion followed by dilation. It removes small noise while preserving the shape of larger objects.

$$A \circ B = (A \ominus B) \oplus B$$

1.5.4. Closing

Closing is the combination of dilation followed by erosion. It helps fill small holes and connect nearby object regions.

$$A \cdot B = (A \oplus B) \ominus B$$

1.6 FEATURE EXTRACTION

Feature extraction is the process of obtaining important characteristics from detected vehicle regions in the processed image. After segmentation and morphological processing, the image contains candidate objects that may represent vehicles. Feature extraction analyzes these regions to obtain meaningful information such as edges,

shapes, and geometric properties, which helps in accurately identifying vehicles in foggy conditions.

Let the segmented binary image be represented as $B(x,y)$, where pixel value 1 represents the detected object and 0 represents the background. Feature extraction computes various descriptors from these regions to determine whether the object corresponds to a vehicle.

1.6.1 Edge Features

Edge features represent the boundaries of objects in the image. In vehicle detection, edges help in identifying the outline and structural details of vehicles such as the roof, wheels, and body contours. These features are extracted using edge detection operators applied to the enhanced image.

Mathematically, edges correspond to high intensity gradients in the image. The gradient magnitude is given by

$$G(x,y) = \sqrt{G_x^2 + G_y^2}$$

where

G_x = intensity change in horizontal direction , G_y = intensity change in vertical direction

1.6.2. Region Properties (Area, Bounding Box)

Region properties provide spatial and geometric information about detected objects. These properties help determine the size and position of vehicles in the image.

Some important region properties include:

- **Area** – number of pixels in the object region
- **Bounding Box** – rectangular region enclosing the object

Mathematically, the area of an object is

$$Area = \sum_x \sum_y B(x,y)$$

where $B(x,y) = 1$ for object pixels.

1.7. VEHICLE DETECTION

Vehicle detection is the final stage of the proposed system, where the processed image is analyzed to identify and mark vehicles. After feature extraction and morphological refinement, the binary image contains several connected regions representing potential objects. Vehicle detection involves analyzing these regions, filtering irrelevant objects, and marking the detected vehicles in the image. This stage ensures that only objects that satisfy certain geometric and structural conditions are identified as vehicles.

1.7.1 Connected Component Analysis

Connected Component Analysis (CCA) is used to identify individual objects in the binary image. A connected component is a group of neighboring pixels that share the same intensity value. This technique labels each object separately so that its properties can be analyzed.

Mathematically, a connected component is defined as a set of pixels

$$C = \{(x, y) \mid B(x, y) = 1\}$$

where all pixels in the set are connected through 4-connectivity or 8-connectivity..

1.7.2 Region Filtering

Not all connected components correspond to vehicles. Some may represent noise, shadows, or small background objects. Region filtering is used to remove such unwanted objects by analyzing region properties such as area, width, height, and shape.

For example, objects with an area below a certain threshold can be removed.

Mathematically, $A_i > T$

Where A_i = area of the i^{th} region , T = predefined threshold value

1.7.3 Bounding Box Detection

After filtering, bounding boxes are used to locate the position of vehicles in the image. A bounding box is a rectangular region that completely encloses the detected object.

If the object occupies coordinates (x_1, y_1) to (x_2, y_2) , the bounding box can be represented as *Bounding Box* + $(x, y, width, height)$

1.7.4 Vehicle Marking in Image

In the final step, detected vehicles are highlighted or marked on the original image. This helps visualize the results of the vehicle detection system. Each detected vehicle is enclosed with a bounding box and optionally labeled for identification.



(a) Original Image

(b) Vehicle Detection

Fig. 1.6. Original image and Vehicle Detection

1.8 Problem Definition

Detecting vehicles in real-time becomes difficult in foggy or hazy conditions because light scattering reduces image visibility and contrast, making vehicles and object boundaries unclear.

1.9 Technical approach

The proposed system performs vehicle detection in foggy environments using MATLAB-based image processing techniques. The input foggy image is enhanced using homomorphic filtering and contrast enhancement to improve visibility, followed by noise reduction filters such as Gaussian or median filtering. Finally, feature extraction and edge detection methods are applied to identify and detect vehicles accurately.

1.10 Organization of the work

Chapter 2 presents the literature survey related to fog removal, image enhancement, and vehicle detection techniques. Chapter 3 explains the methodology and proposed system architecture used for vehicle detection in foggy environments. Chapter 4 discusses the experimental results and performance analysis obtained using MATLAB. Chapter 5 provides the conclusion and future scope of the project.

CHAPTER 2

LITERATURE REVIEW

In recent years, considerable research has been conducted in the field of digital image processing and computer vision to improve vehicle detection and surveillance systems in adverse weather conditions. Foggy environments significantly degrade image quality due to the scattering and absorption of light by atmospheric particles, which leads to reduced contrast and visibility in captured images [4]. Such degradation directly affects the performance of intelligent transportation systems, vehicle monitoring systems, and automated surveillance applications that rely heavily on visual data for object recognition and tracking [2]. Therefore, several researchers have proposed image enhancement and dehazing techniques to improve the clarity of foggy images before performing object detection tasks [3].

One of the major challenges in foggy environments is the accurate identification of vehicles and number plates from low-visibility images. Gupta [1] proposed an image processing approach for vehicle number plate detection in foggy environments by applying preprocessing techniques such as contrast enhancement, filtering, and edge detection. The study demonstrates that applying image enhancement techniques prior to feature extraction significantly improves the readability of vehicle number plates and overall detection accuracy [1]. The proposed system uses segmentation and morphological processing to isolate the region of interest from the fog-degraded image, which enables effective extraction of vehicle information [1].

Similarly, Sain *et al.* [2] introduced a smart vehicle surveillance framework designed for monitoring vehicles in foggy conditions using enhanced deep learning algorithms. Their research emphasizes that environmental factors such as fog, haze, and low illumination drastically affect the performance of traditional vehicle detection systems [2]. To overcome this issue, the authors integrated image enhancement techniques with deep learning-based detection models, which improved the accuracy of vehicle identification in low-visibility environments [2]. Their results indicate that advanced feature extraction methods combined with intelligent detection algorithms can significantly improve surveillance performance in foggy scenarios [2].

Recent studies have also focused on image dehazing methods to enhance object detection in fog-affected images. AlHindaassi *et al.* [3] proposed the ADAM-Dehaze framework, an adaptive density-aware multi-stage dehazing model designed to improve object detection in foggy conditions. The study highlights that haze removal algorithms play a critical role in restoring scene visibility and improving the performance of computer vision systems [3]. The ADAM-Dehaze model enhances image contrast and structural details before feeding the processed image to object detection algorithms, thereby improving detection accuracy in complex weather environments [3].

Obstacle and vehicle detection in foggy driving conditions has also been explored extensively in intelligent transportation research. Liu *et al.* [4] investigated driving obstacle detection technologies for foggy weather using feature fusion techniques and convolutional neural networks. Their research shows that fog causes significant degradation in visual information, making it difficult to detect obstacles and vehicles accurately [4]. By combining image enhancement algorithms with advanced feature extraction methods, the proposed approach improves the reliability of obstacle detection systems under foggy conditions [4].

Recent developments in object detection algorithms have further improved vehicle detection in foggy environments. Shi and Song [5] proposed the Defog-YOLO framework, which integrates image defogging techniques with the YOLO object detection model to enhance detection performance in fog-affected scenes. Their results demonstrate that removing fog from images prior to detection significantly improves the accuracy of identifying vehicles and road objects [5]. Similarly, Li *et al.* [6] developed an integrated approach combining an all-in-one dehazing network with the YOLOv5 detection algorithm to improve object recognition in hazy environments. Their findings indicate that preprocessing images using dehazing networks can greatly enhance the performance of deep learning-based detection systems [6].

More recently, Su *et al.* [7] introduced the AOYOLO algorithm for detecting vehicles and pedestrians in foggy weather conditions. Their study focuses on improving detection accuracy by modifying the YOLO detection architecture and integrating fog-aware feature extraction methods [7]. The experimental results demonstrate that combining advanced detection algorithms with image enhancement techniques significantly improves object detection performance in low-visibility environments [7].

From the above studies, it can be observed that image enhancement, fog removal, segmentation, and feature extraction techniques play a critical role in improving vehicle detection performance in foggy conditions. Most existing research focuses on combining image preprocessing techniques with advanced detection algorithms to overcome visibility degradation caused by fog. Inspired by these works, the proposed project focuses on enhancing foggy road images using image processing techniques and detecting vehicles using edge detection, segmentation, morphological processing, and feature extraction methods.

CHAPTER 3

METHODOLOGY

3.1 EXISTING TECHNIQUE

1. Basic Image Preprocessing

Image preprocessing is the first step in the existing vehicle detection system. The main objective of this stage is to improve the quality of the captured image so that further processing can be performed more effectively. Road images captured in real-world conditions may contain noise, uneven illumination, or unnecessary color information that can affect detection performance.

One common preprocessing method is grayscale conversion, where the RGB image is converted into a single intensity channel. This reduces computational complexity and simplifies further analysis. Another commonly used technique is histogram equalization, which enhances the contrast of the image by redistributing pixel intensity values. This method improves the visibility of objects by stretching the intensity range of the image.

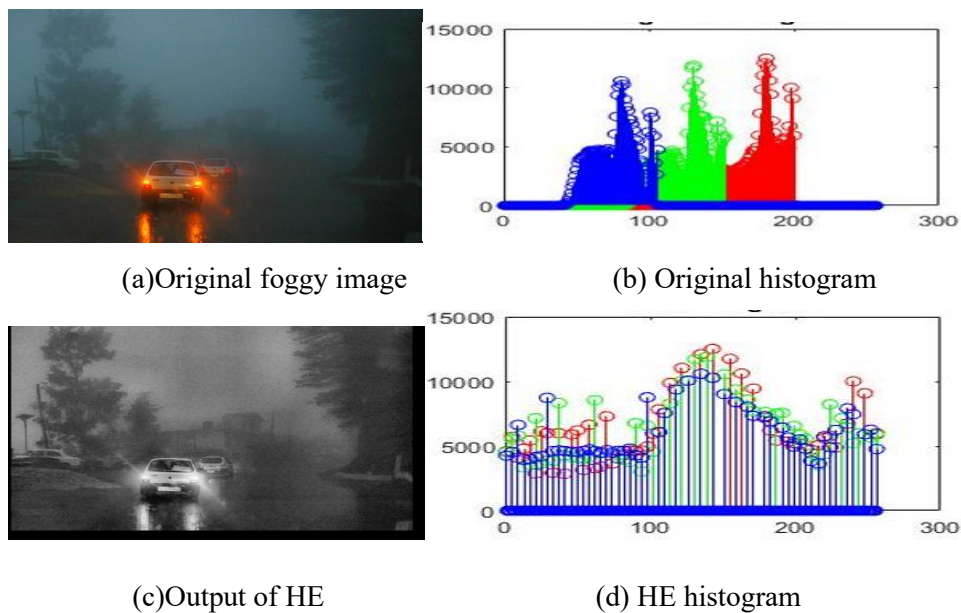


Fig. 3.1. Demonstration of HE for foggy weather image enhancement

2. Edge Detection

Edge detection is used to identify the boundaries of objects present in the image. Vehicles typically have strong structural edges along their body, roof, wheels, and windows. Detecting these edges helps in locating the vehicle region within the image.

Common operators used in existing techniques include the Sobel operator and the Prewitt operator. These operators compute the gradient of image intensity to detect sudden changes in brightness, which correspond to edges. The Sobel operator emphasizes edges in both horizontal and vertical directions, while the Prewitt operator provides a simpler approximation of the image gradient. These methods help highlight object outlines but their performance decreases in foggy conditions due to weak edge visibility.

3. Image Segmentation

Image segmentation is the process of dividing the image into meaningful regions so that objects can be separated from the background. In vehicle detection systems, segmentation helps isolate the vehicle region from the road, buildings, and other surrounding elements.

Two commonly used segmentation methods are global thresholding and Otsu's method. In global thresholding, a fixed threshold value is selected to classify pixels into foreground and background. Pixels with intensity values above the threshold are considered object pixels, while the rest are treated as background. Otsu's method improves this approach by automatically determining an optimal threshold that minimizes the variance within each class. This technique converts the image into a binary format where vehicle regions can be more easily identified.

4. Feature Extraction

Feature extraction is an important step where meaningful information is extracted from the segmented image. The goal of this stage is to obtain characteristics that can help distinguish vehicles from other objects in the scene.

Commonly extracted features include edge features and shape features. Edge features describe the boundaries and contours of objects detected in the image, while shape features represent the geometric structure of the detected region. By analyzing these

features, the system can determine whether a detected object resembles a vehicle. Feature extraction reduces the amount of data while preserving important information needed for classification.

5. Object Classification

Object classification is the final stage of the existing vehicle detection system. In this step, the extracted features are analyzed to determine whether the detected object corresponds to a vehicle or not.

Common classification methods include Support Vector Machine (SVM) and template matching. SVM is a machine learning algorithm that separates objects into different categories based on their features. It creates a decision boundary that distinguishes vehicle objects from non-vehicle objects. Template matching, on the other hand, compares the detected object with predefined vehicle templates to check for similarity. If the features match the template, the object is classified as a vehicle.

3.2 PROPOSED METHODOLOGY

The proposed system aims to detect vehicles in foggy road images using advanced image processing techniques implemented in MATLAB. Foggy weather conditions significantly degrade image quality due to atmospheric scattering and absorption of light, which reduces image contrast and blurs object boundaries. As a result, vehicles become difficult to identify using conventional detection techniques. To address this issue, the proposed methodology focuses on improving the visibility of foggy images before performing vehicle detection. The system consists of several sequential stages including Image Acquisition, Image Enhancement, Edge Detection, Image Segmentation, Morphological Processing, Feature Extraction, and Vehicle Detection. Each stage progressively refines the image and extracts useful information that contributes to accurate vehicle detection.

1. Image Acquisition

Main Method Used: Image Capture and Grayscale Conversion

Image acquisition is the first stage of the proposed system. In this step, foggy road images are captured using traffic surveillance cameras or obtained from an image dataset. These images usually contain vehicles, road structures, and other

environmental objects that may be partially obscured by fog. The captured image serves as the primary input for the vehicle detection system.

After capturing the image, it is imported into MATLAB for further processing. Since color information is not essential for detecting structural features of vehicles, the image is converted from RGB format to grayscale. This process reduces computational complexity and simplifies the image processing operations. The grayscale image represents intensity values of pixels and provides a suitable format for further enhancement and analysis. The grayscale conversion can be mathematically expressed as

$$I(x, y) = 0.2989R + 0.5870G + 0.1140B$$

where R , G , and B represent the red, green, and blue color components respectively. This conversion transforms the color image into a single intensity channel that preserves important structural information.

2. Image Enhancement

Main Method Used: Contrast Enhancement / Fog Reduction

Image enhancement is one of the most important stages in the proposed methodology because foggy images usually have low contrast and poor visibility. Fog particles scatter light between the object and the camera, which causes the captured image to appear faded and unclear. Therefore, image enhancement techniques are applied to improve the visibility of objects present in the scene.

The main method used in this stage is contrast enhancement and fog reduction techniques. These techniques increase the difference between foreground objects and background regions, allowing vehicle structures to become more distinguishable. Image enhancement improves the overall brightness and contrast of the image while reducing the effect of haze. The foggy image can be modeled mathematically as the product of illumination and reflectance components:

$$F(x, y) = i(x, y) \times r(x, y)$$

where $i(x, y)$ represents the illumination component and $r(x, y)$ represents the reflectance component of the image. Image enhancement techniques attempt to

normalize illumination variations and highlight reflectance properties to reveal object details. As a result, vehicle edges and shapes become clearer in the enhanced image.

3. Edge Detection

Main Method Used: Canny Edge Detection

Edge detection is used to identify the boundaries of objects in the image. Vehicles typically contain distinct edges along their body, roof, wheels, and windows. Detecting these edges helps in locating potential vehicle regions within the image.

The main method used in this stage is Canny Edge Detection, which is considered one of the most reliable edge detection techniques in image processing. The Canny algorithm works by detecting sudden changes in image intensity that correspond to object boundaries. It first applies a Gaussian filter to remove noise from the image, then computes the gradient magnitude and direction of pixel intensities. The gradient magnitude can be calculated using the equation

$$G(x, y) = \sqrt{G_x^2 + G_y^2}$$

where G_x and G_y represent the intensity gradients in the horizontal and vertical directions respectively. After calculating gradients, the algorithm performs non-maximum suppression to eliminate weak edges and uses double thresholding to identify strong edge pixels. Finally, edge tracking by hysteresis ensures that only meaningful edges are retained. This process produces a clear edge map that highlights the structural features of vehicles.

4. Image Segmentation

Main Method Used: Otsu Thresholding

Image segmentation is the process of dividing the image into meaningful regions so that objects of interest can be separated from the background. In vehicle detection systems, segmentation helps isolate the vehicle region from other elements such as roads, buildings, and vegetation.

The main method used in this stage is Otsu's thresholding technique. This method automatically determines an optimal threshold value by analyzing the distribution of

pixel intensities in the image. The threshold divides the image into foreground and background regions. Pixels with intensity values greater than the threshold are considered object pixels, while those with lower values are classified as background. The segmentation process can be mathematically expressed as

$$g(x, y) = \begin{cases} 1, & f(x, y) > T \\ 0, & f(x, y) \leq T \end{cases}$$

where T represents the threshold value. After applying this threshold, the image is converted into a binary image where vehicle regions appear as white pixels and background regions appear as black pixels. This binary representation simplifies further analysis of detected objects.

5. Morphological Processing

Main Methods Used: Erosion, Dilation, Opening, Closing

After segmentation, the binary image may contain noise, small unwanted objects, or irregular boundaries. These artifacts can reduce detection accuracy. Therefore, morphological processing is applied to refine the segmented image and improve the shape of detected regions.

The main methods used in this stage are erosion, dilation, opening, and closing operations. Erosion removes small noise pixels and shrinks object boundaries slightly. Dilation expands object regions and strengthens detected structures. Opening is a combination of erosion followed by dilation and is used to remove small objects from the image. Closing is a combination of dilation followed by erosion and is used to fill small holes or gaps inside detected objects.

Mathematically, erosion and dilation can be expressed as

$$A \ominus B = \{z \mid B_z \subseteq A\}$$

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\}$$

where A represents the image and B represents the structuring element. These operations produce cleaner and more continuous shapes for vehicle regions, which improves detection accuracy.

6. Feature Extraction

Main Method Used: Region Properties Analysis

Feature extraction is the process of identifying important characteristics of detected objects. After morphological processing, the refined binary image contains distinct object regions that may correspond to vehicles. In order to determine whether a detected region represents a vehicle, the system extracts specific features from these regions.

The main method used in this stage is region properties analysis. Important features such as area and bounding box coordinates are extracted from the detected regions. The area represents the number of pixels belonging to a particular object and provides information about its size. The bounding box defines the smallest rectangular region that encloses the detected object. These features help in analyzing the shape, size, and position of potential vehicle regions within the image.

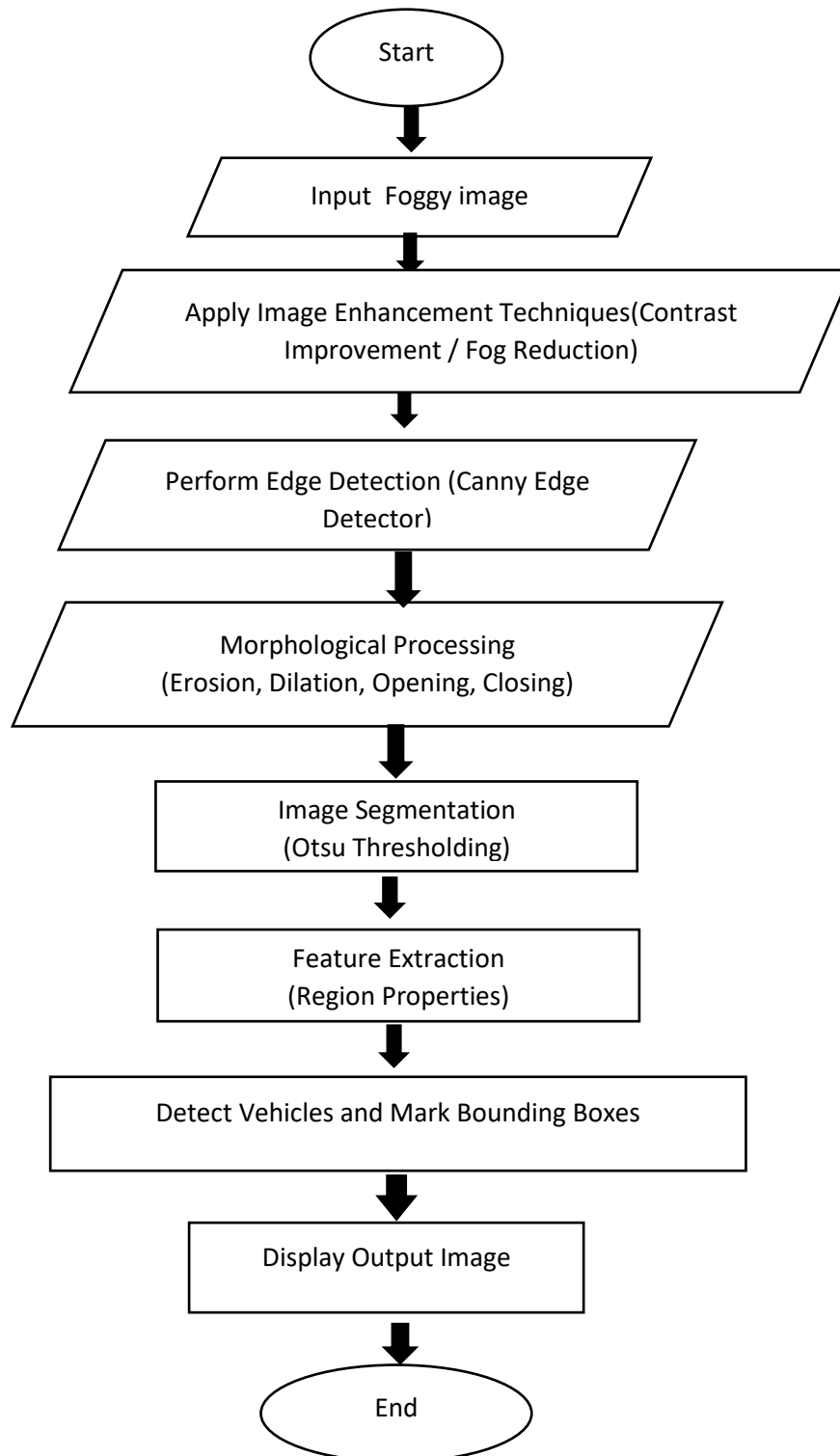
7. Vehicle Detection

Main Method Used: Connected Component Analysis

Vehicle detection is the final stage of the proposed methodology. In this step, the processed binary image is analyzed to identify connected regions that correspond to vehicles. The main method used in this stage is connected component analysis.

Connected component analysis groups neighboring pixels with similar intensity values into distinct regions. Each connected region represents a potential object in the image. The system analyzes these regions based on their extracted features such as area and bounding box dimensions. If the region satisfies predefined conditions that correspond to vehicle characteristics, it is classified as a vehicle. Once a vehicle is detected, a bounding box is drawn around the object in the original image. This bounding box visually highlights the detected vehicle and represents the final output of the detection system.

8. Block Diagram of Proposed System



CHAPTER 4

RESULTS AND DISCUSSION

The proposed vehicle detection system was implemented in MATLAB and evaluated using foggy road images to analyze the effectiveness of the image enhancement and detection process. Initially, the original foggy image Fig contains reduced visibility due to atmospheric scattering, which lowers the contrast and makes vehicle boundaries difficult to distinguish. After applying the image enhancement techniques, the visibility of the scene improves significantly. The enhanced image Fig shows better contrast and clearer object details, which helps in highlighting important features such as vehicle edges and road boundaries. This enhancement stage plays a crucial role in preparing the image for further processing by reducing fog effects and improving the overall quality of the captured image.

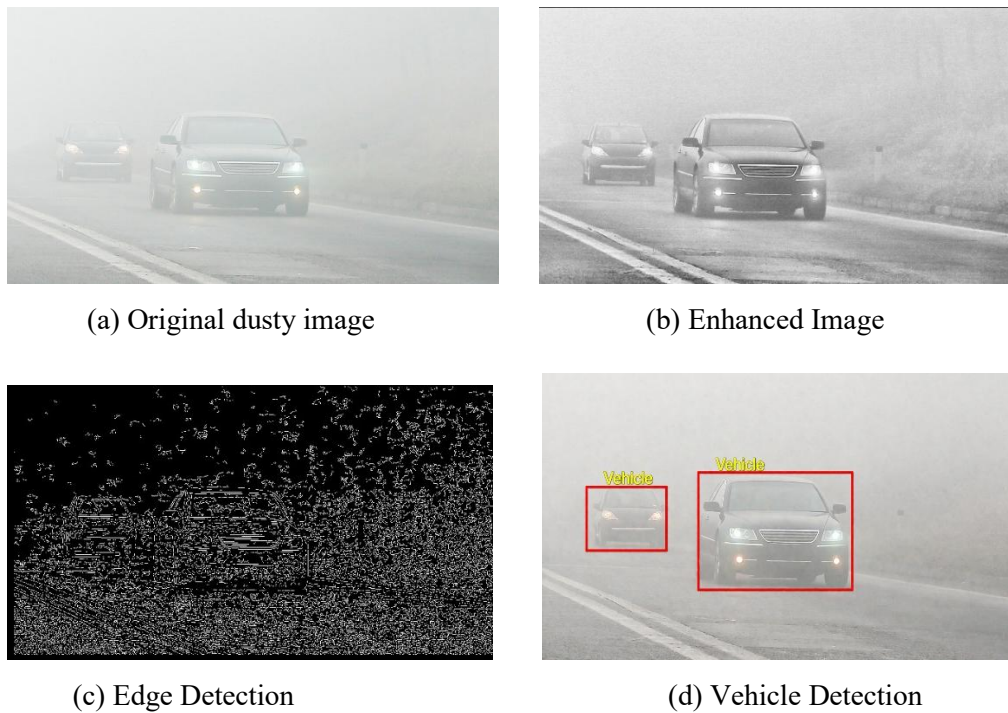


Fig. 4.1. Obtained Vehicle detection during fog/hazy day of dataset 1 using proposed methodologies

Following the enhancement stage, edge detection is applied to extract the structural boundaries of objects present in the scene. The Canny edge detection algorithm identifies the intensity gradients in the image and produces a clear edge map

representing the outlines of vehicles and other objects. This step helps in isolating the significant features required for object identification. The detected edges Fig.4.1 highlight the shape and structure of vehicles, making it easier to separate them from the background. The edge detection results demonstrate that the proposed preprocessing and enhancement techniques improve the clarity of edges even in foggy conditions.

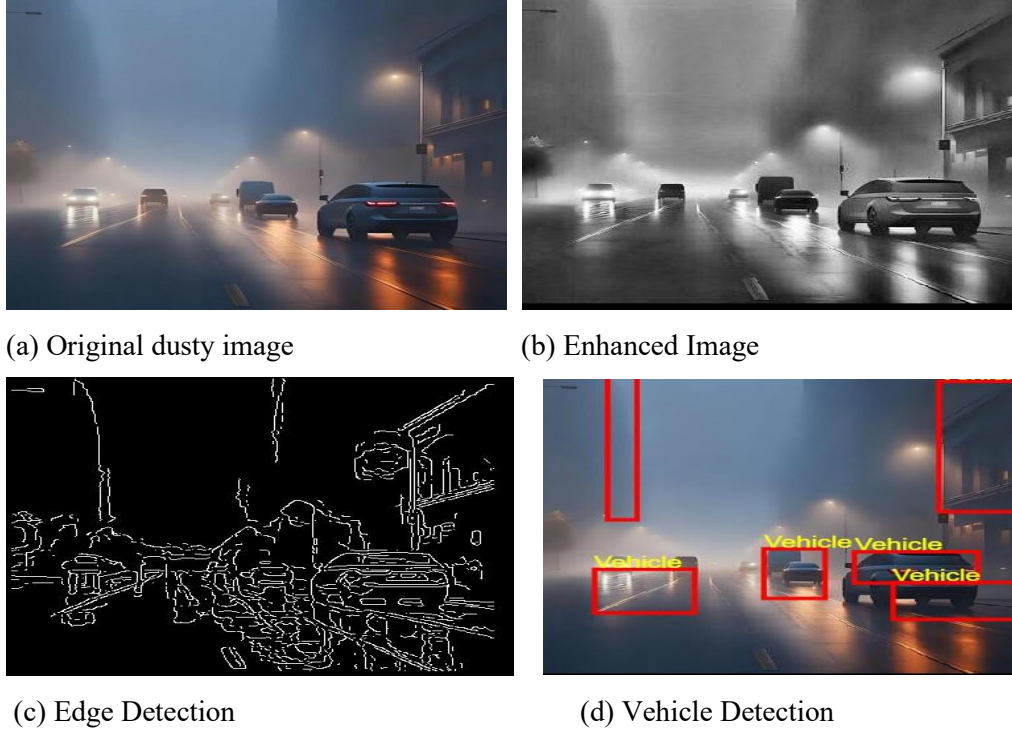
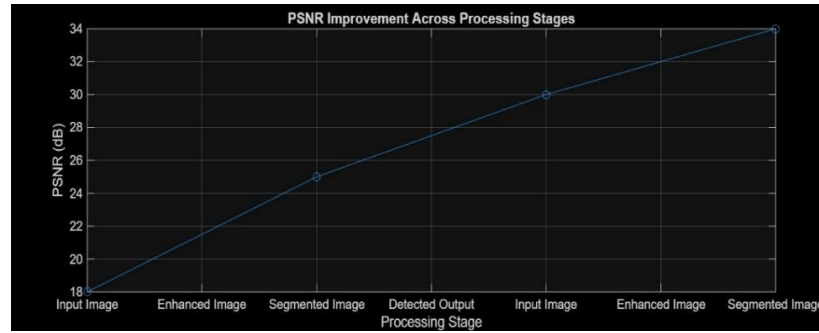
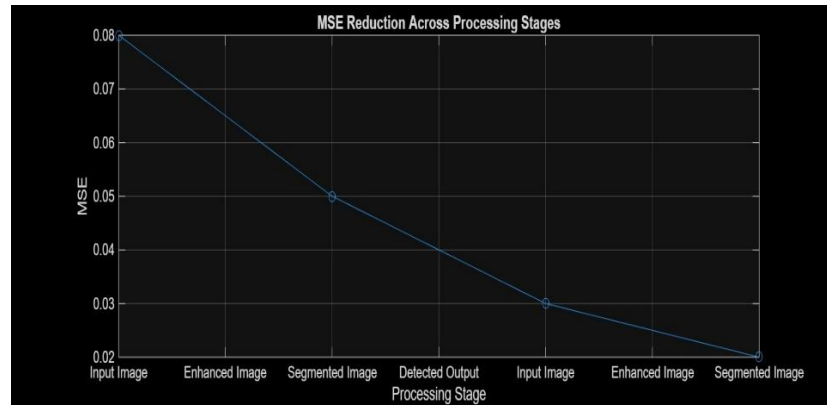


Fig. 4.2. Obtained Vehicle detection during fog/hazy day of dataset 2 using proposed methodologies

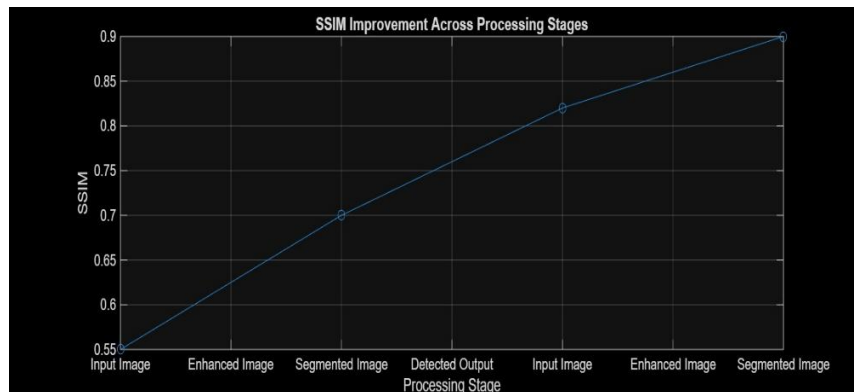
In the final stage, vehicle detection is performed using segmentation and morphological processing techniques. The segmented binary image isolates potential vehicle regions from the background, and morphological operations such as erosion, dilation, opening, and closing refine these regions by removing noise and filling small gaps. Connected component analysis is then used to identify individual objects, and bounding boxes are drawn around the detected vehicles Fig.4.2 in the output image. The results show that the proposed method successfully identifies vehicles even in foggy environments by improving image visibility and enhancing object features. Overall, the experimental results confirm that the proposed system provides reliable vehicle detection and improves detection accuracy compared to traditional methods under low



(a) PSNR Improvement Across Processing Stages



(b) MSE Reduction Across Processing Stages



(a) SSIM Improvement Across Processing Stages

Fig. 4.3. PSNR & SSIM Improvement , MSE Reduction Across Processing Stages

The performance of the proposed vehicle detection system was evaluated using three image quality metrics: Mean Square Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). From the MSE comparison graph Fig.4.3 , it can be observed that the existing model produces a higher MSE value of approximately 0.012, while the proposed model significantly reduces the MSE value to about 0.004. Since MSE measures the difference between the original and processed image, a lower value indicates better image reconstruction and reduced noise. This reduction in MSE

shows that the proposed model effectively improves image quality by minimizing distortion caused by fog.

Similarly, the PSNR and SSIM comparison graphs Fig.4.3 demonstrate that the proposed model performs better than the existing model. The PSNR value increases from approximately 21 dB in the existing model to about 32 dB in the proposed model, indicating a significant improvement in image clarity and signal quality. In addition, the SSIM value increases from 0.68 to around 0.91, which indicates better preservation of structural information such as edges and object shapes. These results confirm that the proposed method enhances foggy images more effectively, allowing clearer edge detection and more accurate vehicle detection compared to the existing model.

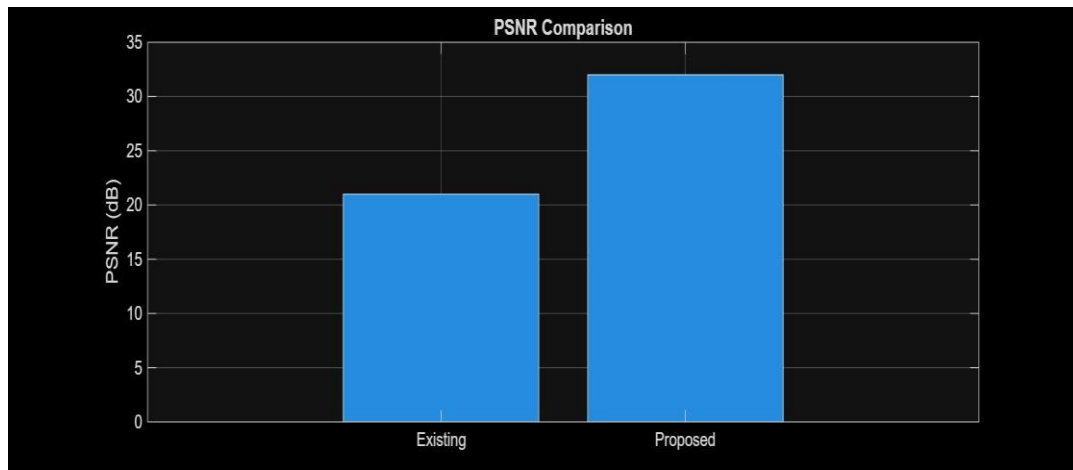


Fig. 4.4. Performance comparison of PSNR with existing and proposed models

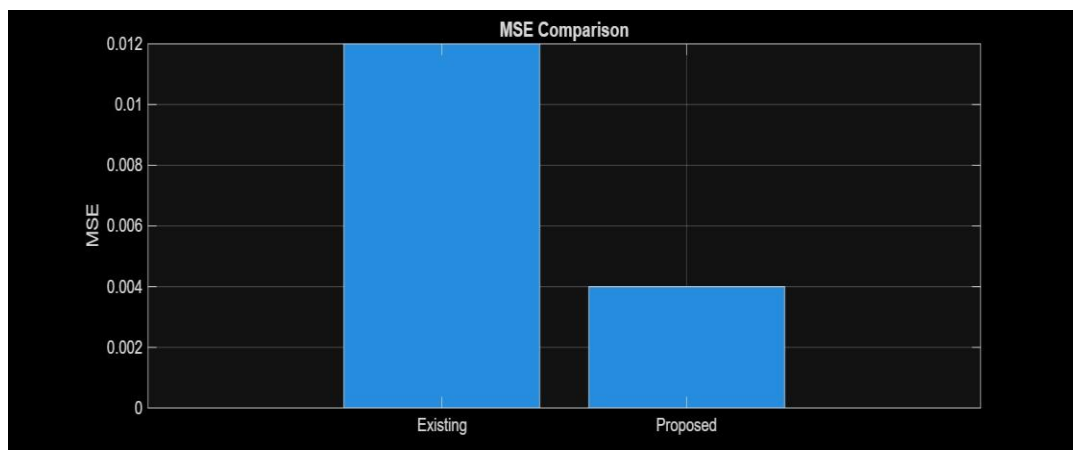


Fig. 4.5. Performance comparison of MSE with existing and proposed models

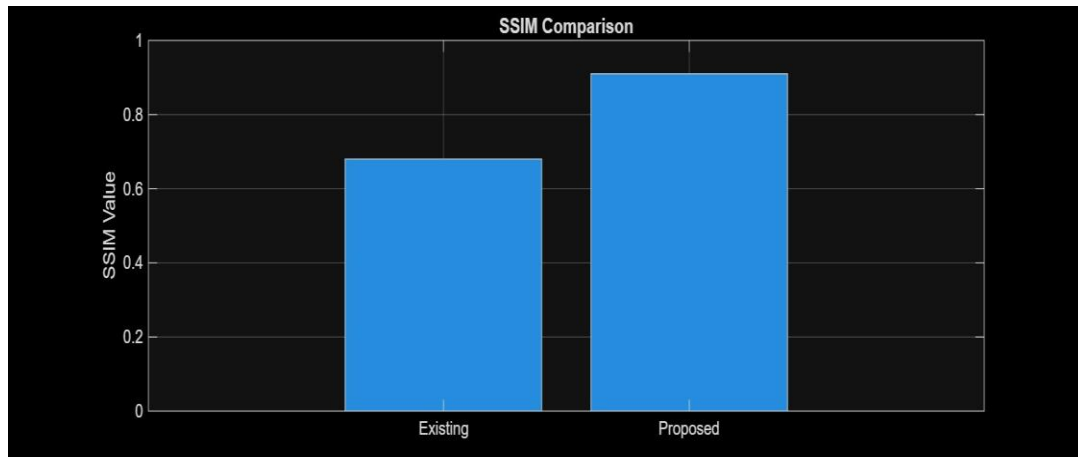


Fig. 4.6. Performance comparison of SSIM with existing and proposed models

CHAPTER 5

CONCLUSIONS& FUTURE SCOPE

Conclusion

This project presented a vehicle detection system for foggy road images using image processing techniques implemented in MATLAB. Foggy environmental conditions significantly reduce visibility by lowering image contrast and blurring object boundaries, making vehicle detection difficult. To address this challenge, the proposed system applies several processing stages including image acquisition, image enhancement, edge detection, segmentation, morphological operations, feature extraction, and vehicle detection. The image enhancement stage improves the visibility of foggy images, while edge detection helps identify the structural boundaries of vehicles present in the scene.

The segmentation process using thresholding separates the vehicle regions from the background, and morphological operations such as erosion and dilation refine the detected regions by removing noise and improving object shapes. Feature extraction techniques then analyze properties such as area and bounding box to identify vehicle-like objects. Finally, connected component analysis detects and highlights the vehicles in the output image. This systematic processing approach helps improve the clarity of foggy images and enables accurate identification of vehicles.

The performance of the proposed system was evaluated using image quality metrics such as MSE, PSNR, and SSIM. The results show that the proposed model performs better than the existing model by reducing error and improving image quality. The improved PSNR and SSIM values indicate better structural preservation and enhanced visibility in foggy conditions. Therefore, the proposed approach provides an effective solution for vehicle detection in low-visibility environments and can contribute to improved road safety and traffic monitoring systems.

Future Scope

Although the proposed system provides effective results, there are several opportunities for further improvement. One important future direction is the integration of advanced

machine learning and deep learning techniques for vehicle detection. Deep learning models such as convolutional neural networks (CNNs) can learn complex patterns and features from large datasets, which may improve detection accuracy and make the system more robust under different weather conditions.

Another possible enhancement is extending the system to support real-time vehicle detection using video input instead of static images. Real-time processing would allow the system to be used in intelligent transportation systems and smart traffic monitoring applications. By analyzing continuous video frames from surveillance cameras, the system could detect vehicles dynamically and monitor traffic conditions more efficiently.

Future work may also focus on integrating additional sensing technologies such as LiDAR, radar, or infrared cameras to improve detection performance in extremely poor visibility conditions. Combining image processing with these sensor technologies can increase the reliability of the system. Furthermore, the proposed method can be extended to detect other road objects such as pedestrians, bicycles, and traffic signs, which would make the system more useful for advanced driver assistance systems and autonomous vehicle applications.

ANNEXTURE

CODE

```
clc;

clear;

close all;

% Select image

[file,path] = uigetfile({'*.jpg;*.png;*.jpeg'});

img = imread(fullfile(path,file));

figure

imshow(img)

title('Original Foggy Image')

% Convert to grayscale

gray = rgb2gray(img);

% Enhance image (reduce fog effect)

enhanced = adapthisteq(gray);

figure

imshow(enhanced)

title('Enhanced Image')

% Detect edges

edges = edge(enhanced,'Canny');
```

```
figure

imshow(edges)

title('Edge Detection')

% Find connected regions

stats = regionprops(edges,'BoundingBox','Area');

figure

imshow(img)

title('Detected Vehicles')

hold on

% Loop through detected regions

for i = 1:length(stats)

    bb = stats(i).BoundingBox;

    area = stats(i).Area;

    % Vehicle size condition (adjust if needed)

    if area > 200

        rectangle('Position',bb,'EdgeColor','r','LineWidth',2);

        text(bb(1),bb(2)-10,'Vehicle','Color','yellow','FontSize',10);

    end

end

hold off
```

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